

Total No. of Questions : 8]

[Total No. of Printed Pages : 3

Roll No

AU/ME-225-CBCS

B.E. IV Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Fluid Mechanics

Time : Three Hours

Maximum Marks : 60

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume data suitably.

1. a) Define with mathematical expression:
 - i) Uniform and Non-uniform flow
 - ii) Steady and unsteady flow
 - iii) One, two and three dimensional flow
- b) State Newton's law of viscosity. Explain the importance of viscosity in fluid motion.
2. a) Explain the following terms:
 - i) Path line
 - ii) Streak line
 - iii) Stream line
 - iv) Equipotential line
- b) Derive Euler's equation for flow along stream line and deduce the Bernoulli's Equation for the same.

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[2]

3. a) The inlet and throat diameters of horizontal venturimeter are 30cm and 10cm respectively. The liquid flowing through the meter is water. The pressure intensity at inlet is 13.734 kN/m^2 while vacuum pressure head at the throat is 37cm of mercury. Find the rate of flow. Assume that 4% of the differential heads is lost between the inlet and outlet. Find also value of C_d for venturimeter.
- b) Derive an expression for continuity equation.
4. a) Discharge Q of a centrifugal pump can be assumed to be dependent on density of liquid ρ , viscosity of liquid μ , pressure, impeller diameter D and speed N in R.P.M. Using Buckingham π -Theorem, show that:

$$Q = ND^3 \phi \left[\frac{gH}{N^2 D^2}, \frac{V}{ND^2} \right]$$

- b) What is drag and lift? Explain different types of drag on immersed body.
5. a) Find the head lost due to friction in a pipe of diameter of 300mm and length 50m, through which water is flowing at a velocity of 3m/s using Darcy equation and Chezy's formula for which $L = 60$. Given kinematic viscosity 0.01 stores.
- b) Derive an expression of velocity and shear stress distribution for laminar flow between fixed parallel plates.
6. a) Define Reynold's Number? What is its significance? List the characteristics of Laminar flow.
- b) Derive an expression for Total Pressure Force and depth of centre of pressure for a vertical surface submerged in water.

[3]

7.
 - a) Explain the different types of similitude.
 - b) Define the following terms and write relevant equations for the same:
 - i) Stagnation temperature
 - ii) Stagnation pressure
8. Write notes on:
 - i) Momentum Thickness
 - ii) Sub sonic and Super sonic flow
 - iii) Buoyancy
 - iv) Surface Tension
